PROBLEM FORMULATION, ENVIRONMENT CLASSIFICATION & STATE TYPE

RAILWISE: *RAILWAY PLATFORM ALLOCATION AI AGENT*

1. Problem Formulation

RailWise evaluates train requirements and station conditions to recommend a feasible platform for each required time window. Its decisions must respect platform compatibility, occupancy, closures, and other restrictions. If no valid option exists, the agent flags the conflict for railway staff instead of forcing an assignment.

Element	Description
Initial State	The agent receives the timetable, train identities and platform requirements, current platform occupancy, existing assignments, known closures, restrictions, and reported delays.
Actions	Update station information; generate candidate platforms; check compatibility and time-window availability; rule out closed or restricted platforms; recommend or revise an assignment; and alert staff when no valid option exists.
Transition Model	When arrivals, departures, delays, closures, occupancy, or assignments change, the station-state model is updated. The set of feasible platforms and the recommended allocation may therefore change.
Path Cost	Among valid assignments, reduce waiting time, timetable deviation, and unnecessary reallocations. Hard constraints—such as compatibility, closures, and non-overlapping platform use—cannot be traded away to reduce cost.
Goal Test	A compatible platform is recommended for the train's required time window without overlapping use or violating restrictions. If no valid platform is available, the case is explicitly flagged for staff review.
State Space	All possible station conditions represented by train statuses and time windows, platform occupancy, existing assignments, closures, restrictions, delays, and proposed allocation decisions.
Search Space	Possible sequences of updating station information, generating candidate platforms, applying constraints, comparing feasible choices, making recommendations, and rechecking after new events.
Agents	RailWise is the intelligent decision-support agent. Railway staff, train crews, and operational systems also affect the wider environment, but RailWise does not control signals or authorize train movements.

2. Environmental Properties

Property	Classification	Reason
Observability	Partially Observable	The agent depends on the station data it receives. Reports may be delayed or incomplete, so it may not immediately know every train or platform condition.
Determinism	Stochastic / non-deterministic	Delays, equipment problems, and operational changes can alter the expected timetable and platform availability.
Episodes	Sequential	One train's platform assignment affects which platforms remain available for later trains; decisions must account for earlier allocations.
Change	Dynamic	Occupancy, arrival times, departures, closures, and restrictions may change while the agent is operating.
Values	Discrete & Continuous	Platform identities and train assignments are discrete, while arrival times, release times, waiting periods, and delays are numerical.
Agents	Multi-agent environment	Railway staff, train crews, and operational systems also influence the environment, although RailWise itself is one decision-support agent.

3. Choosing the State Type

The state type describes how the agent represents the situation it is solving. RailWise must maintain a model of changing train and platform conditions, compare possible allocations, and revise its recommendation when new information arrives.

Option	Suitability	Explanation
Single-state	Not suitable	A single snapshot cannot represent changing occupancy, multiple trains, different time windows, and disruptions that affect the next allocation.
Multi-state	Most suitable	The agent moves among many possible station states: current timetable and occupancy, candidate platform sets, feasible allocations, updated schedules, and conflict states requiring staff review.

CONCLUSION
RailWise is best represented as a **multi-state** problem because it evaluates many possible combinations of train schedules, platform availability, restrictions, and allocation choices. It is also **partially observable** and **stochastic**: the agent relies on available station information, while unexpected delays or operational changes can alter the situation. Its planning is sequential because each allocation can affect later trains.